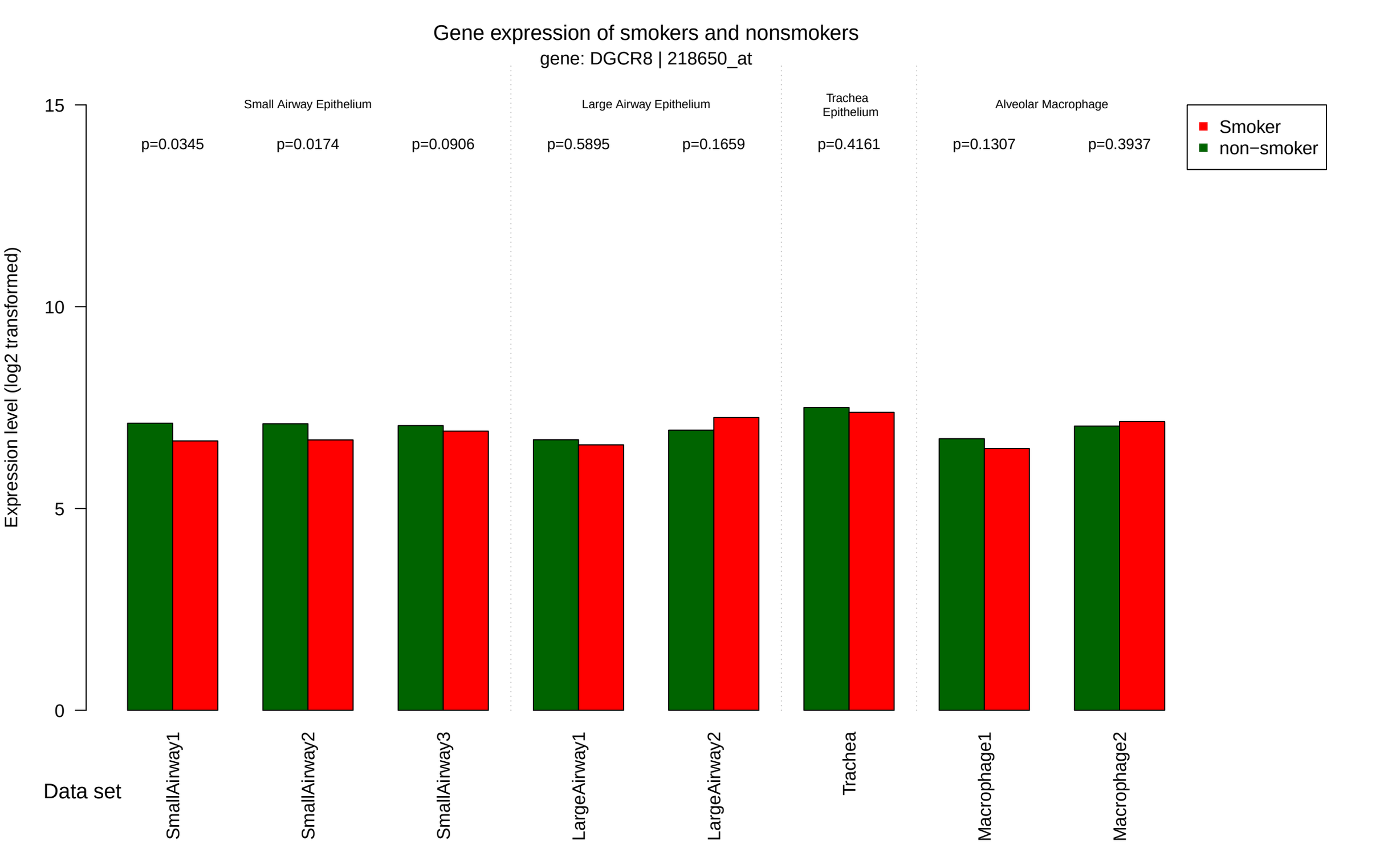
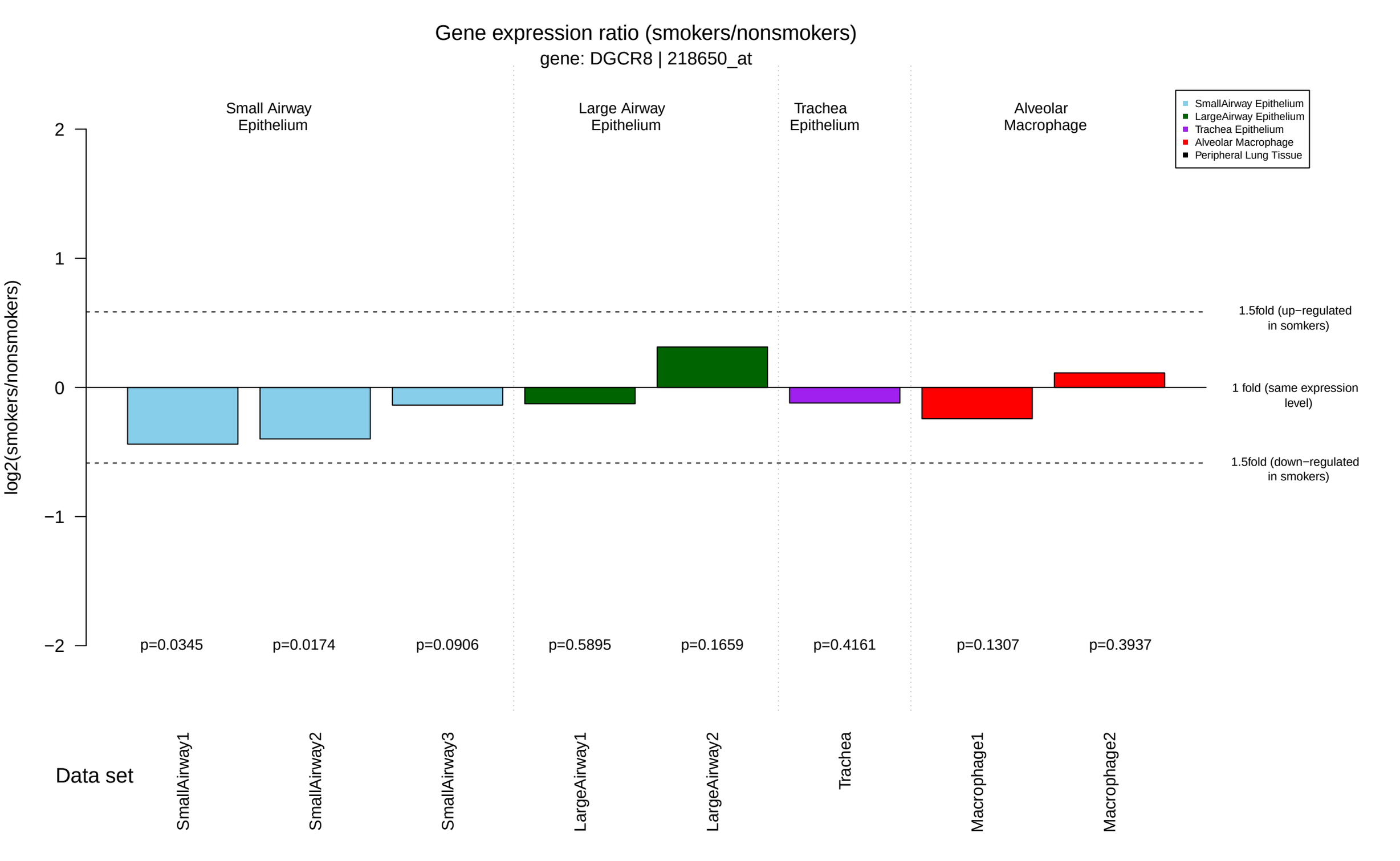
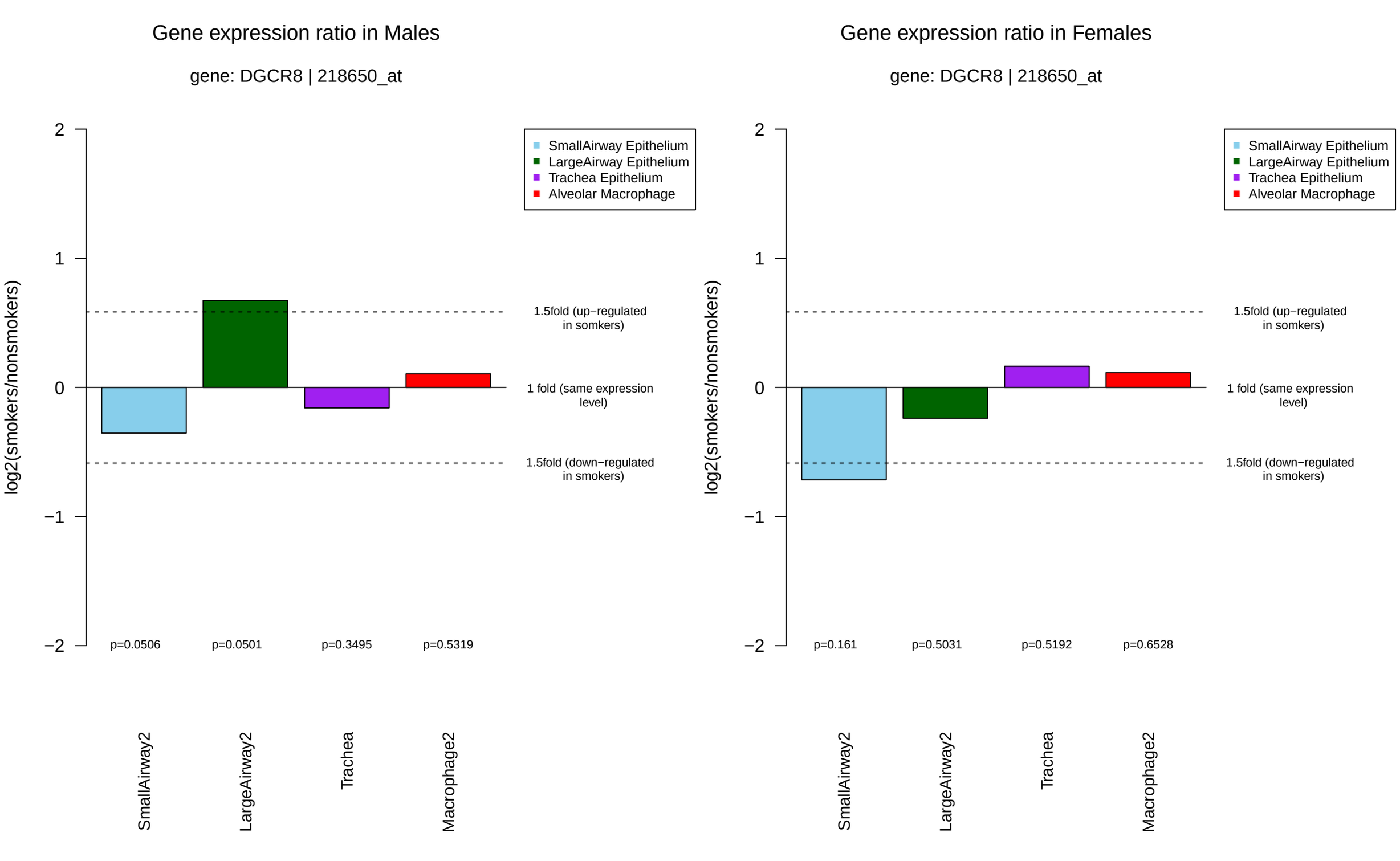
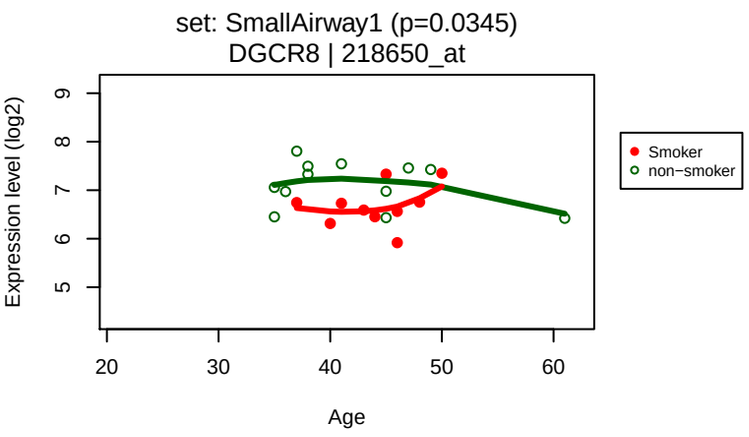
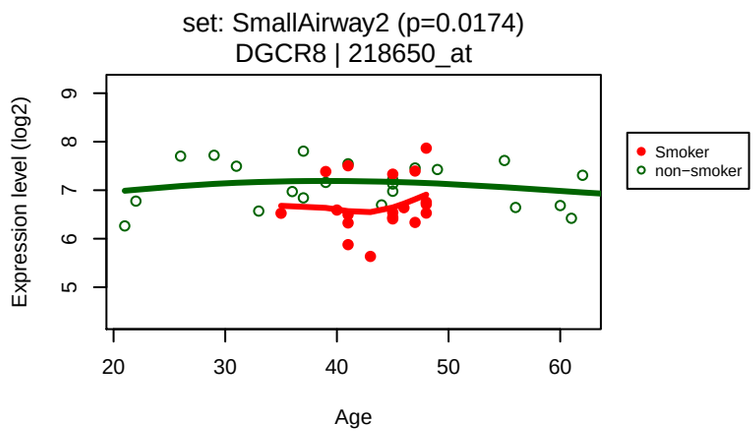
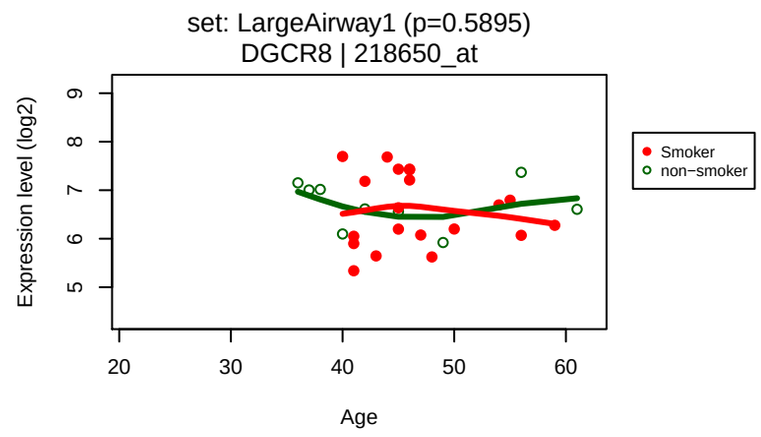

Gene expression of smokers and nonsmokers
gene: DGCR8 | 218650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0345
p=0.0174
p=0.0906
p=0.5895
p=0.1659
p=0.4161
p=0.1307
p=0.3937
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

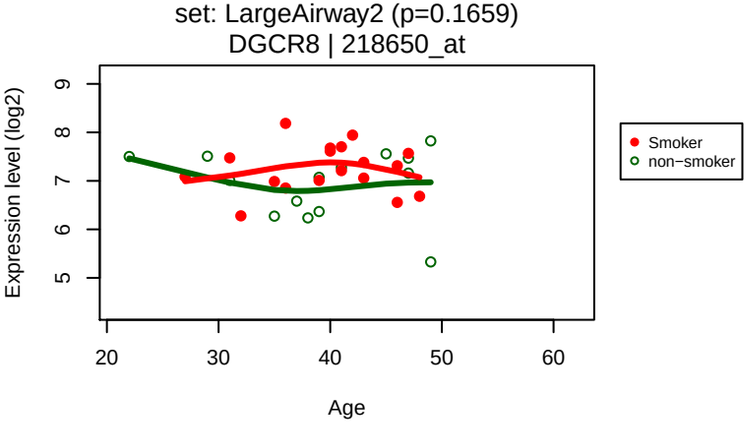

Gene expression ratio (smokers/nonsmokers)
gene: DGCR8 | 218650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0345
p=0.0174
p=0.0906
p=0.5895
p=0.1659
p=0.4161
p=0.1307
p=0.3937
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

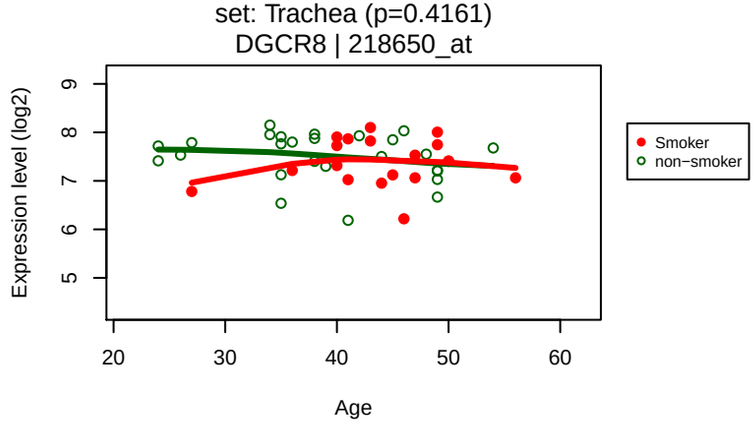

Gene expression ratio in Males
gene: DGCR8 | 218650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0506
p=0.0501
p=0.3495
p=0.5319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DGCR8 | 218650_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.161
p=0.5031
p=0.5192
p=0.6528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

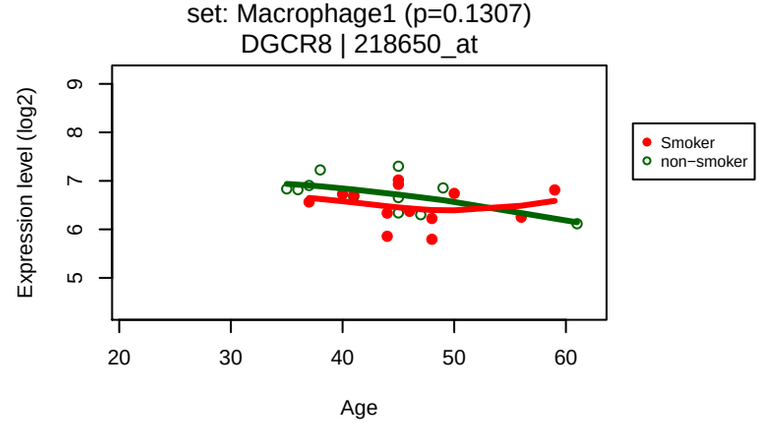

set: SmallAirway1 (p=0.0345)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker

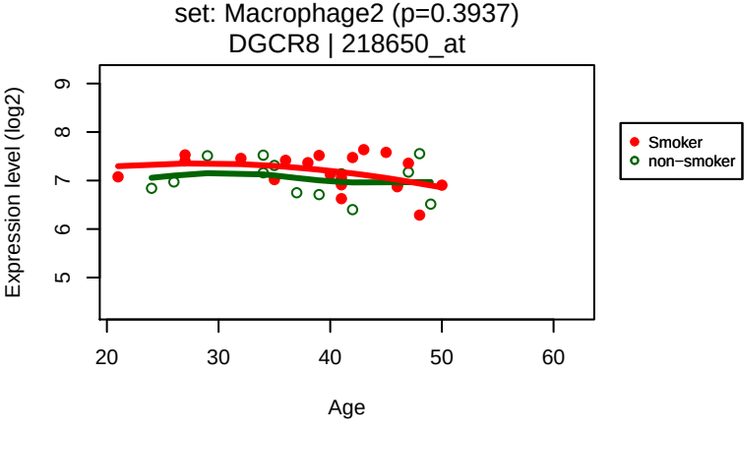

set: SmallAirway2 (p=0.0174)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker

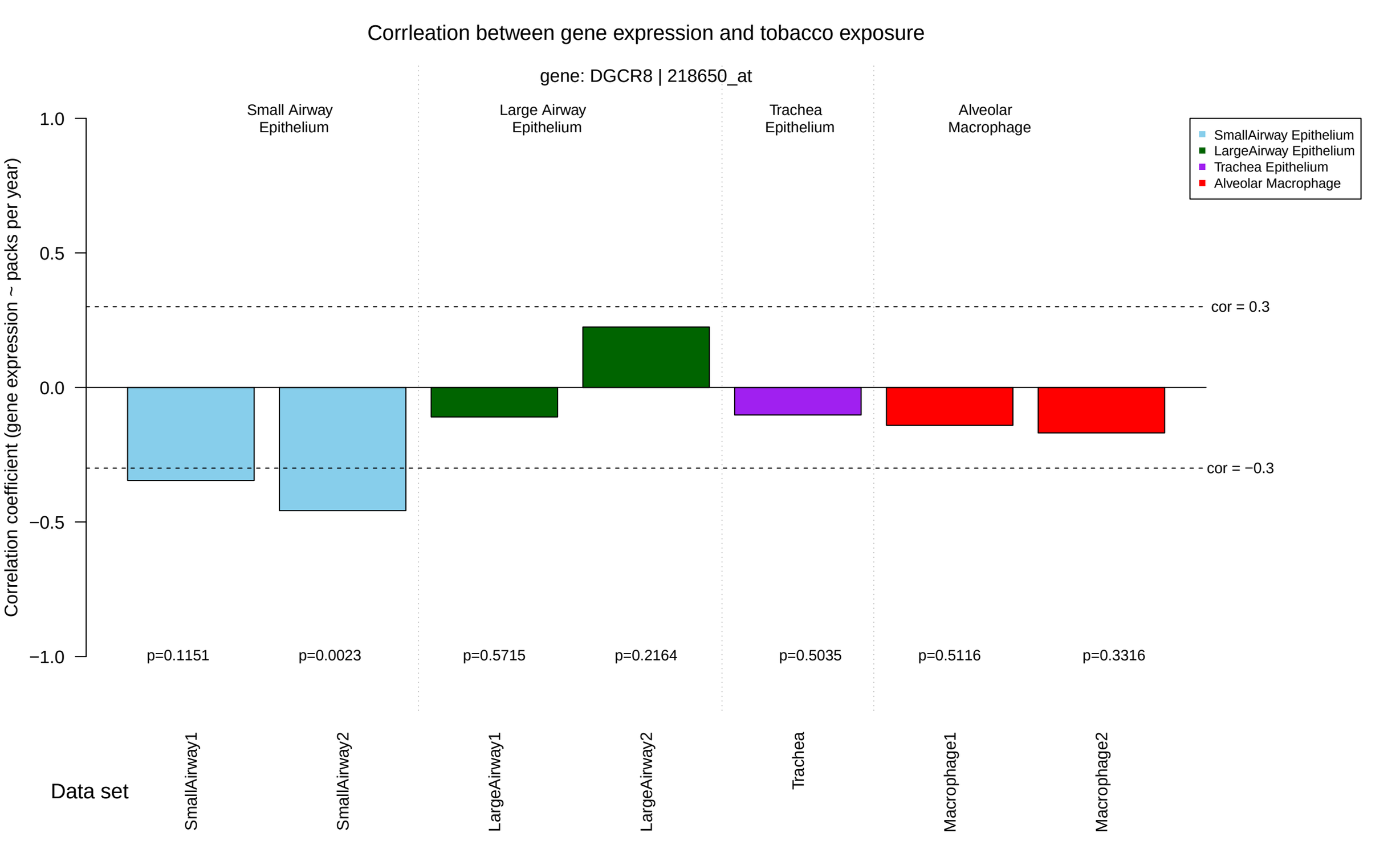

set: LargeAirway1 (p=0.5895)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1659)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4161)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1307)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3937)
DGCR8 | 218650_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DGCR8 | 218650_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1151
p=0.0023
p=0.5715
p=0.2164
p=0.5035
p=0.5116
p=0.3316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2